Chapter 6: t-Tests

Statistics for Environmental Science

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Where Student's *t* came from

In the early 1900s, William Sealy Gosset worked as a brewer at Guinness, where quality control depended on testing ingredients such as barley and hops. Every test used up beer or ingredients, so Guinness could only test small samples. The statistics of the time relied on the normal (z) distribution, which works when samples are large.

Gosset set out to measure how much extra uncertainty small samples add. He derived the **Student's t distribution**, the sampling distribution of t when the population standard deviation is estimated from the sample. Guinness restricted outside publication, so he published the 1908 paper under the pseudonym "Student" (Student 1908), which is why we still say Student's t-test. This chapter uses Gosset's approach. We measure an effect, divide it by its standard error, and compare the resulting t to the t distribution for $n - 1$ degrees of freedom (defined later in this chapter).

We often compare a sample mean to a reference (drinking-water standard, historical average) or compare two conditions (before/after, site A vs site B). A t-test asks whether the observed difference is larger than we would expect from sampling variation alone, given n and the variability in the data. Chapter 5 added two questions to every test: how large the effect is, and whether the design had the power to detect it. Both carry over to the t-tests here.

Three common t-tests you will use.

1. **One-sample t-test.** Compare a sample mean to a target (μ_0).
2. **Paired t-test.** Compare before/after (or matched) measurements on the same units by running a one-sample t-test on the *differences*.
3. **Independent-samples t-test (also called two-sample t-test).** Compare means from two groups measured on different units. The pooled version, explained later in this chapter, assumes the two groups have similar spread. Welch's version (Welch 1947) does not assume this. R's `t.test()` runs Welch's version unless you set `var.equal = TRUE`.

From *z* to *t*

A test statistic is the effect you observed divided by an estimate of how much it could vary by chance:

$$\text{statistic} = \frac{\text{effect}}{\text{error}}$$

In Chapters 4 and 5 we used this with z to test a sample mean against μ_0:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

The numerator is the effect, the distance between the sample mean and the value H_0 names. The denominator is the standard error, the standard deviation of the sampling distribution of the sample mean. Means vary less than individual observations do, by a factor of $\sqrt{n}$, which is why the denominator is $\sigma/\sqrt{n}$ rather than σ.

z needs the population standard deviation σ, and real studies almost never know it. The t-test estimates it with the sample standard deviation s, the spread of the estimated population distribution from Chapter 4, so the denominator is the estimated standard error from the end of Chapter 3, $s/\sqrt{n}$:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

t still measures the distance from μ_0 in standard errors. The difference is that s is itself an estimate that changes from sample to sample, and the rest of this section shows what that does to the distribution of t.

Why *t* has heavier tails than *z*

Because s varies from sample to sample, the t distribution has heavier tails than the standard normal. Heavier tails means more of the curve's area is far from zero, so large values happen more often for t than for z. This happens because some samples have an s well below σ, which shrinks the denominator and makes t larger. Figure 1 shows how much s varies at two sample sizes. As n grows, s stabilizes and t converges to z.

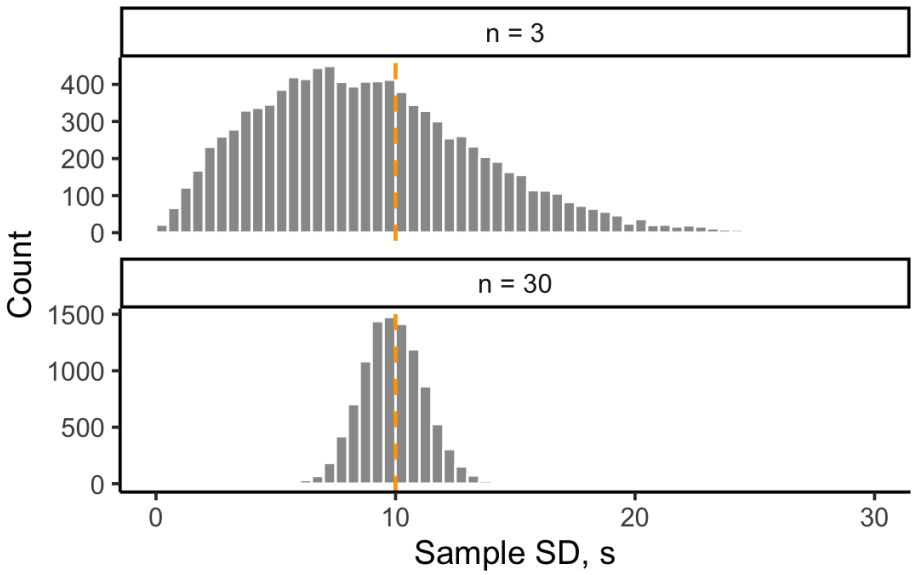


Figure 1: Simulated sample SDs from 10,000 samples of each size, all drawn from one normal population with $\sigma = 10$ (orange dashed line). With $n = 3$, s is often far below or above 10. With $n = 30$, it stays close to 10.

Figure 2 plots the normal curve with the t curves for 2 and 10 degrees of freedom. In the top panel the curves look almost the same. The bottom panel puts the vertical axis on a log scale, which makes the small tail probabilities easier to see, and there the t curve for 2 degrees of freedom sits well above the normal curve in both tails.

> Tip: When is z appropriate?
>
> If the population SD σ is known *and* the sampling distribution of the sample mean is (approximately) normal, use z. In practice, σ is rarely known, so we use t with s and $df = n - 1$ (degrees of freedom, defined in the next section).

Degrees of freedom

Figure 2 drew a different t curve for each value of the **degrees of freedom**, df. The degrees of freedom are the number of values still free to vary once something has been estimated from the data. The df is also the number in parentheses whenever a t-test is reported, as in $t(9) = 1.94$.

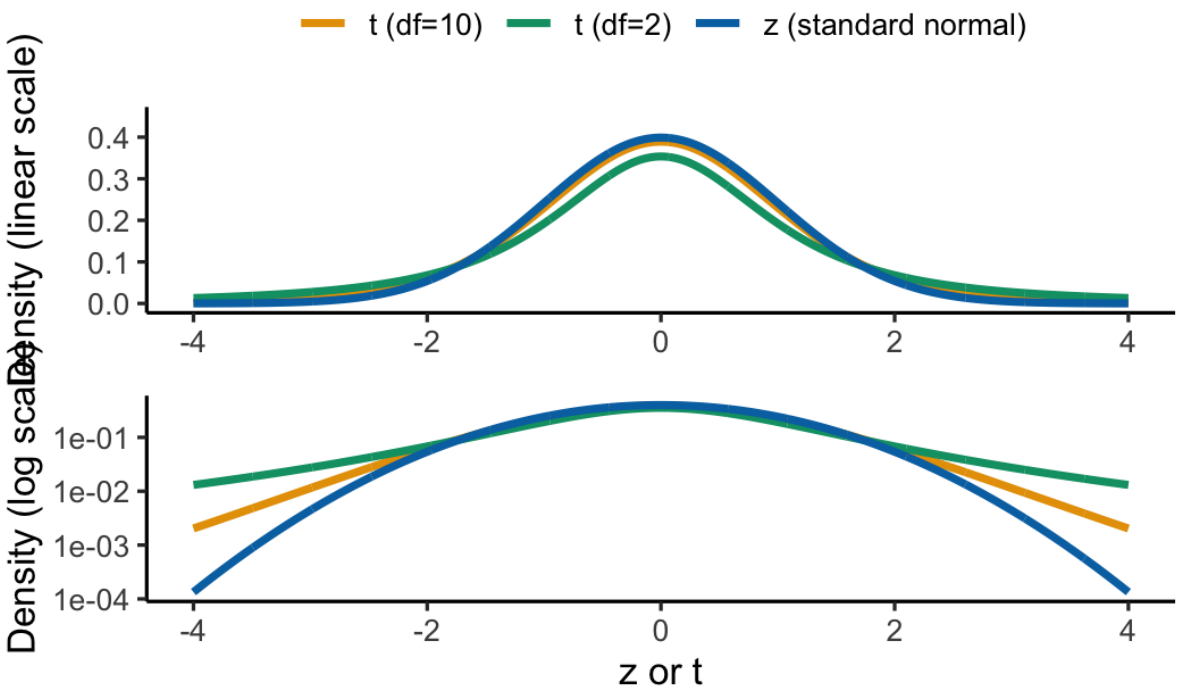


Figure 2: Normal (z) and t distributions. On a linear scale (top), the tail differences look small. On a log scale (bottom), the heavier tails of t with few degrees of freedom are visible.

For one-sample and paired t-tests, $df = n - 1$. A sample of 10 has $df = 9$.

Consider three numbers with a mean of 2. Three numbers with a mean of 2 must add up to $3 \times 2 = 6$. Pick the first two freely, say 51 and -3. They add up to 48, so the third number has to be

$$x_3 = 6 - (51 + (-3)) = 6 - 48 = -42.$$

Two of the three numbers were free to vary, and the third was not. That's $df = n - 1 = 2$ for three numbers.

In a t-test, the mean is the thing estimated, and the df tells you how much information is left for estimating the spread, s. With few degrees of freedom, s can be well above or below σ, so the t curve has the heavier tails described earlier in this chapter. As df grows, s settles close to σ and the t curve approaches the normal. Each df therefore gets its own t curve.

In a simulation with H_0 true, 7.8% of t values from samples of 10 fall beyond ± 2, against 4.9% from samples of 101, close to the 5% a normal curve gives. A test with few degrees of freedom therefore needs a larger cutoff (2.26 at $df = 9$) to keep the Type I error rate at 5%.

Critical values

For two-tailed tests at $\alpha = 0.05$, the critical values (Chapter 4) are:

- z uses critical values ± 1.96.
- t uses wider cutoffs for small samples: ± 2.26 at $df = 9$, ± 2.09 at $df = 20$, ± 2.05 at $df = 29$. They approach ± 1.96 as n increases.

Rule of thumb. Under H_0, $|t|$ beyond about 2 happens about 5% of the time once df is 30 or more.

t when H_0 is true

t is positive when $\bar{x}$ is above μ_0 and negative when $\bar{x}$ is below it. Under H_0, t follows the t distribution with $df = n - 1$.

To see how t behaves when H_0 is true, we can simulate it. We set up a population whose mean really is $\mu_0 = 0$, draw a sample of 10, and compute one t. Then we repeat that 1,000 times.

Before you look. Each bar in Figure 3 counts t values from these samples. Decide roughly what share of them will fall beyond ± 2.26, the critical values for $df = 9$.

Most of the 1,000 values fall between -2 and 2. About 4% fall beyond ± 2.26, close to the 5% the cutoff is built to allow. The blue curve is the t distribution with 9 degrees of freedom, and the histogram follows it. Software uses that curve in place of a simulation to find how often chance alone produces a t as large as the one observed in a study.

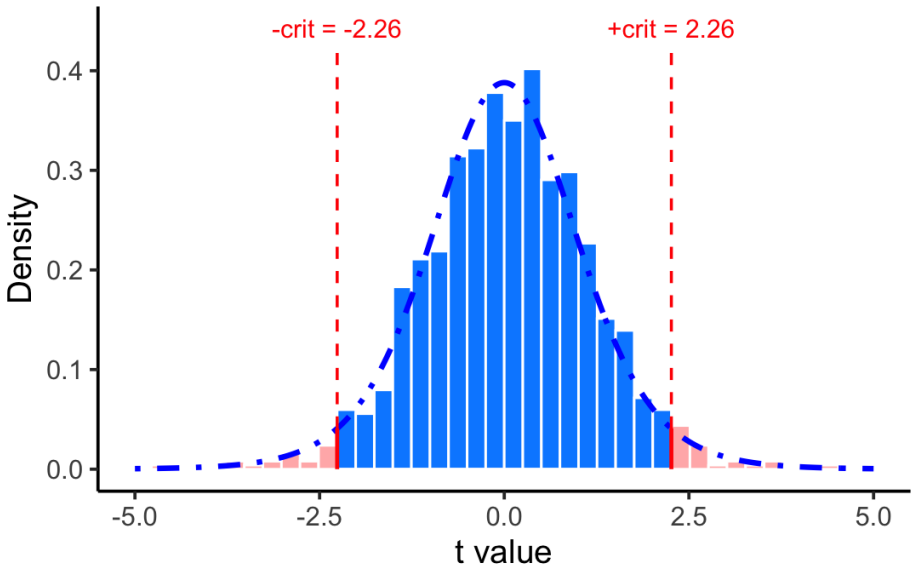


Figure 3: Histogram of 1,000 simulated t values from samples of 10 drawn from a population whose mean equals $\mu_0 = 0$. Bars beyond the critical values, ± 2.26, are shaded red. The blue dot-dash curve is the t distribution with 9 degrees of freedom.

The one-sample *t*-test

Use a one-sample t when you want to test whether a sample mean differs from a reference value (μ_0). It asks whether the observed difference is bigger than you'd expect from sampling variability, given n and your sample SD s.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad df = n - 1$$

After calculating t from our data, we compare the observed t to a t distribution with $n-1$ degrees of freedom to get a p-value and a decision at α.

Calculating *t* from data

Let's calculate a one-sample t statistic step by step using a short example. A new highway opened near a residential area, and a noise guideline for the area sets a daytime level of 50 dB. Ten monitoring sites near the highway each recorded an average daytime sound level, in decibels (dB). The sound levels are invented. They are:

Sound levels (dB): 50, 70, 60, 40, 80, 30, 90, 60, 70, 60

- **Null (H_0):** the mean sound level near the highway equals the 50 dB guideline, $\mu = 50$.
- **Alternative (H_a):** it differs from the guideline, $\mu \neq 50$.

By hand (the squared deviations from 61 add up to 2,890):

$$\bar{x} = 610/10 = 61 \text{ dB}$$

$$s^2 = 2890/9 = 321.1$$

$$s = \sqrt{321.1} = 17.92 \text{ dB}$$

The standard error of the mean is the sample SD divided by the square root of n:

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}} = \frac{17.92}{\sqrt{10}} = 5.67$$

And t is the difference between the sample mean (61 dB) and the reference value (the 50 dB guideline), divided by that standard error:

$$t = \frac{\bar{x} - \mu_0}{SE_{\bar{x}}} = \frac{61 - 50}{5.67} = 1.94$$

We can confirm this with a single R command:

```
t.test(sound_db, mu = 50)
```

Results: $t(9) = 1.94$, $p = 0.084$, mean = 61.0 dB, 95% CI [48.18, 73.82].

Confidence interval. The 95% CI in that output is $\bar{x} \pm t_{crit} \cdot SE_{\bar{x}}$, with t_{crit} taken from the t distribution with $n - 1$ degrees of freedom. Here $t_{crit} = 2.262$, the same two-tailed cutoff for $df = 9$ as in the critical values section, so the interval is $61 \pm 2.262(5.67)$, or [48.18, 73.82]. The interval contains 50, so the test does not reject H_0. For a two-tailed test at α, the $100(1 - \alpha)\%$ CI and the critical value always reach the same decision, the same equivalence Section 4.4.3 showed for z. The critical value is a cutoff on the t scale, and we compare the observed t to it. The confidence interval is a range on the scale of the data (dB here), built around $\bar{x}$, and we check whether it contains μ_0. Both use the same t_{crit} and the same standard error, so they agree.

Interpretation. The mean (61 dB) is above the 50 dB guideline, but the difference is not statistically significant at $\alpha = 0.05$.

Assumptions check. The 10 sound levels spread evenly around the mean of 61 dB with no outliers, so approximate normality is reasonable. Section covers how to check this.

Interpreting *t*

The highway noise t of 1.94 can be placed among the t values that chance alone produces. Under H_0, sample means far from μ_0 are unlikely, so large values of $|t|$ are rare. The decision rule for a t-test follows from this. If the observed t falls beyond the critical values, reject H_0.

The **null distribution** of t is the distribution of t values across many samples when H_0 is true. For the highway noise example, it is the distribution of t values from many samples of 10 sites if the true mean were exactly the 50 dB guideline. The 1,000 simulated t values in Figure 3 are draws from this distribution at $df = 9$. It is the t version of Chapter 4's chance window and of the sampling distribution of the sample mean under H_0. It is a different curve from the null population distribution, which describes individual values. Figure 4 shows what that distribution looks like, at $df = 9$.

Under H_0, t is most likely to be near zero. The distribution is symmetric, so half of all t-values are at or above zero and half are at or below.

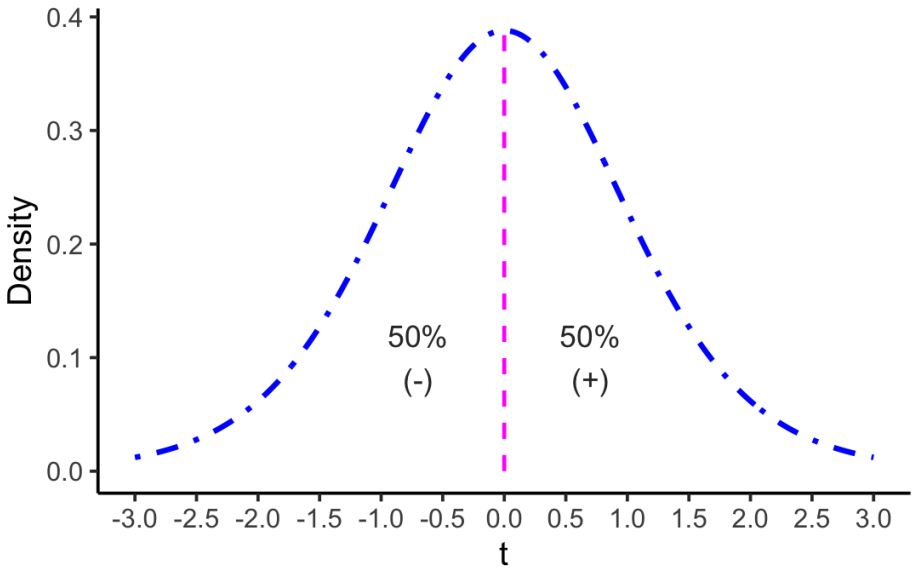


Figure 4: The distribution of t-values that chance alone produces when the population mean equals μ_0 ($df = 9$).

We can get more precise than a 50/50 split. Figure 5 divides the same distribution into ten regions, each containing 10% of the distribution. Notice the regions get wider out in the tails: since most t-values cluster near 0, it takes a wider interval out there to capture the same 10%.

For example:

1. $t \leq -1.38$ contains 10% of the t values.
2. $-1.38 \leq t \leq -0.88$ contains 10% of the t values.
3. $-0.88 \leq t \leq -0.54$ contains 10% of the t values.
4. $t \geq 1.38$ contains 10% of the t values.

The highway noise t of 1.94 is beyond 1.38, so it falls in the top 10% of this distribution. Its exact p-value comes from software, and the one-tailed section below uses it.

Getting *p*-values for *t*

Older textbooks printed p-values for t in a table at the back of the book. Today R, Excel, and online calculators give them directly. You'll practice this in lab, both with software and by simulation.

Software needs two things from you to compute a p-value: the degrees of freedom, which you already have, and whether the test is one-tailed or two-tailed, which comes next.

One-tailed vs. two-tailed, applied

Sections [-4.5.1] and [-5.2.1] covered when each is justified. For a t-test, the tail choice changes both the critical value and the p-value for the exact same data.

For the highway noise example ($df = 9$), a one-tailed test puts the whole 5% rejection region in one tail, so its critical value is 1.83. A two-tailed test splits the 5% into 2.5% in each tail, which pushes the critical values farther out, to ± 2.26.

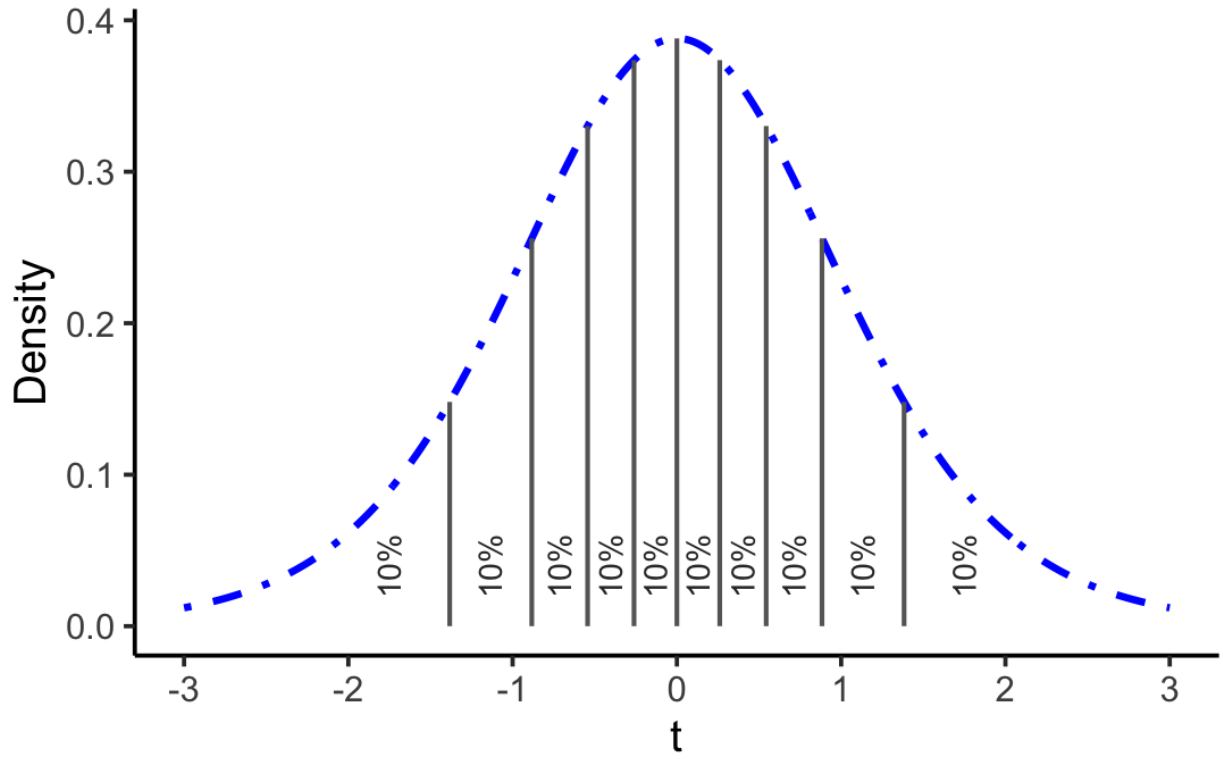


Figure 5: Splitting the t distribution up into regions each containing 10% of the t-values. The width between the bars narrows as they approach the center of the distribution, where there are more t-values.

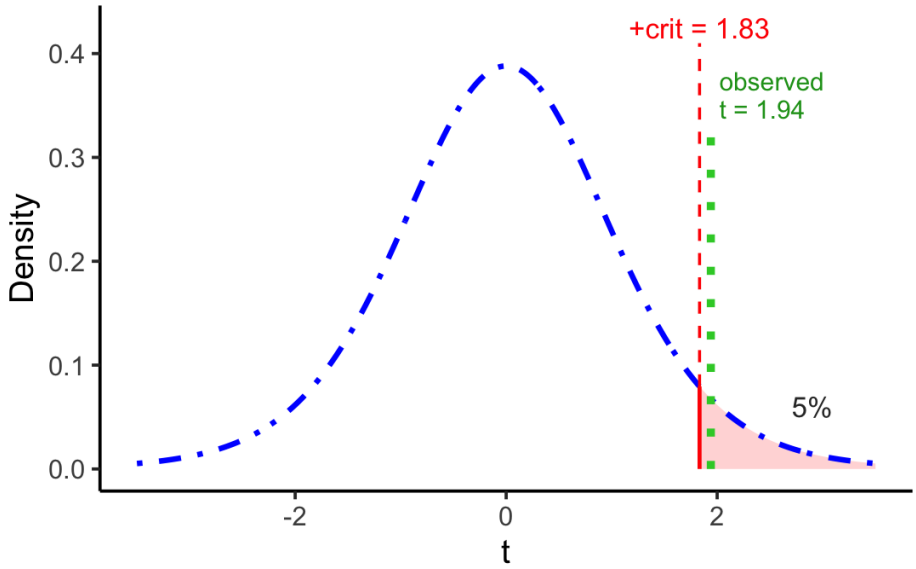


Figure 6: One-tailed critical value at $df = 9$. When H_0 is true, a t value of 1.83 or greater occurs 5% of the time.

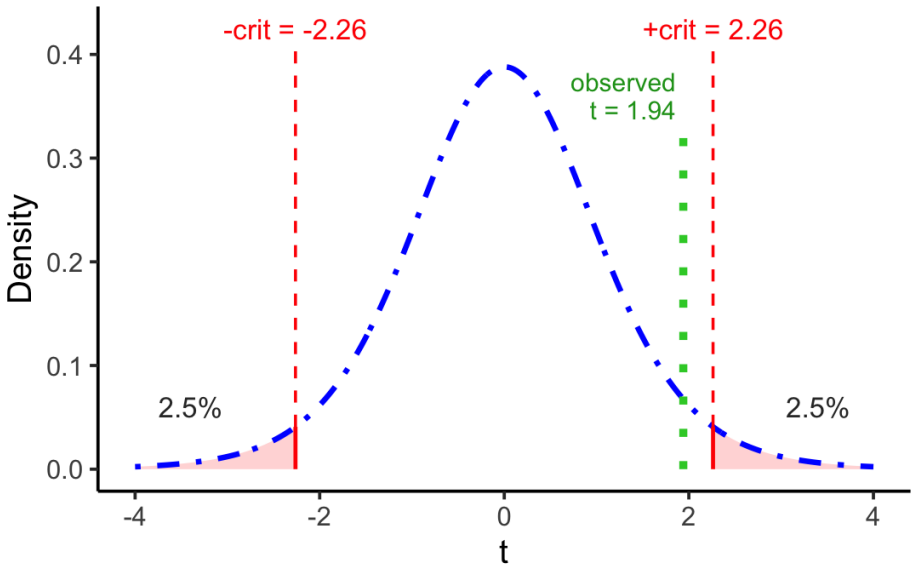


Figure 7: Two-tailed critical values at $df = 9$. When H_0 is true, values of t beyond ± 2.26 occur 5% of the time in total, 2.5% in each tail.

For our result, $t(9) = 1.94$.

Test	H_a	p	At $\alpha = 0.05$
One-tailed	$\mu > 50$	0.042	significant
Two-tailed	$\mu \neq 50$	0.084	not significant

The two-tailed p is exactly double, because a two-tailed test counts the matching region in both tails, and here the choice changes the decision.

Assumptions, and what to do when they don't hold

Every test in this book has assumptions. Assumptions are conditions that need to be roughly true for the math behind the test to be trustworthy. There are two different kinds of assumptions.

The universal assumption. Every test in this book, including the nonparametric tests in Chapter 7, makes two universal assumptions. First, the observations are independent, so knowing one observation's value tells you nothing about another's (the definition in Section 3.5.2, applied to data). In a paired design, the pairs are the observations. Second, the data come from a sound study design, meaning a fair sample that does not systematically favor certain values. Because it applies everywhere, we won't list it test by test from here on. When an exam or homework question asks you to "list the assumptions of this test," it means the test-specific assumptions below.

Test-specific assumptions. These come from how a particular test's math works, and they differ test to test. For the one-sample and paired t-tests, the specific assumption is approximate normality of the outcome itself (one-sample) or of the difference scores (paired). The t distribution we've been using to get p-values is *derived* assuming the population is normal, which makes the sample mean normally distributed too. When the population is far from normal, the p-value may not be the true probability of a result this extreme under H_0.

Whether non-normality matters depends on n. With a large sample, the sampling distribution of the sample mean tends toward normal even when individual observations are skewed (the central limit theorem), so the t-test's p-values stay close to correct. Much environmental field data comes from small samples that are also skewed. Examples include a handful of expensive water samples, a short list of monitoring wells, or the five plots you had time to visit in a season. A t-test's answer becomes questionable when the sample is small and the data are also non-normal.

What to do about it. First, look at your data. A histogram or a Q-Q plot (Section 9.6.2 shows how to read one) will usually tell you whether "approximately normal" is a reasonable description or a stretch. You can also run a formal normality test, the **Shapiro-Wilk test** (`shapiro.test()` in R), which tests H_0: the data came from a normal distribution. A small Shapiro-Wilk p-value is evidence that the data did not come from a normal distribution. A large p-value means the test found no such evidence, which is weaker than showing the data are normal. Treat the result as one more piece of evidence alongside the plot. With a small sample, which is when a violation matters most, the test has low power and often fails to reject H_0 even when the data are far from normal. If your data look far from normal and your sample is small, you can use a nonparametric test instead, which does not assume normality. You'll meet these nonparametric alternatives in the next chapter, once we've covered all three t-test forms below.

Table 1 summarizes the one-sample t-test.

Table 1: Test card: one-sample t-test.

Purpose	Compare a sample mean to a reference value μ_0 For example, a standard, a historical mean, or a policy target
Hypotheses	$H_0 : \mu = \mu_0$ $H_a : \mu \neq \mu_0$
Assumptions	Population approximately normal
Test statistic	$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$ $df = n - 1$
Report	$\bar{x}$ and 95% CI $t(df)$ and p

The paired *t*-test

When the same units are measured twice (before/after, matched pairs), the question is whether they changed. The data for the test are difference scores. A **difference score** is one unit's second measurement minus its first.

Design in this worked example. Ecologists measured soil moisture at 20 field plots before wetland restoration and again one year later, to see whether restoration increased moisture enough to support plant reestablishment and carbon sequestration. The soil moisture values are invented. Figure 8 shows the design.

The dashed boxes mark what makes this design paired. In the design figures in earlier chapters, the boxes held *different* units, and the test compared one set against another. Here both boxes hold the same 20 plots, so each plot's Before value can be subtracted from its own After value.

Define differences and their meaning. Choose an order and stick to it:

- $d_i = \text{After}_i - \text{Before}_i$
- $d_i > 0$: plot had more soil moisture after restoration (increase)
- $d_i < 0$: plot had less soil moisture after restoration (decrease)

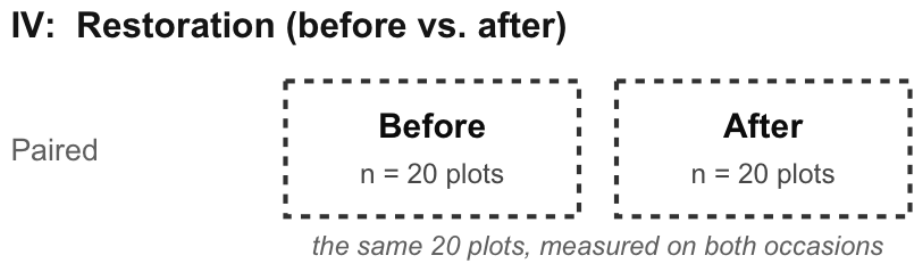


Figure 8: A paired design. The same 20 plots are measured before and after restoration.

This subtraction cancels between-plot variability (some plots are naturally wetter than others) and focuses on within-plot change, so the difference scores vary less than the raw soil moisture values do. A smaller s_d gives a smaller standard error, so the paired test has more power than an independent-samples test (the third test in this chapter) on the same data.

- **Null (H_0):** restoration did not change soil moisture, $\mu_d = 0$, where μ_d is the mean difference score in the population of plots.
- **Alternative (H_a):** restoration changed soil moisture, $\mu_d \neq 0$.

Test on differences (paired t). Compute one difference per plot, then run a one-sample t against 0:

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i, \quad s_d = \mathrm{SD}(d_1, \ldots, d_n), \quad t = \frac{\bar{d} - 0}{s_d/\sqrt{n}}, \quad df = n - 1.$$

Here n is the number of pairs, 20 plots in this study.

t has the same sign as $\bar{d}$. If you subtracted Before from After, a positive t means moisture went up. Reversing the order flips the sign of every d_i, $\bar{d}$, and t, and leaves p unchanged.

What to report. $\bar{d}$ (the mean difference score), a 95% CI, $t(df)$, and p. Optionally add a standardized effect size, $\bar{d}/s_d$, the paired version of Cohen's d. Cohen's d is the effect size from Section 5.3.

Table 2 separates the two uses of the letter d.

Table 2: The two uses of the letter d.

Symbol	Meaning
d_i, $\bar{d}$, s_d	Each plot's difference score (After minus Before), the mean of the difference scores, and their SD
Cohen's d	The effect size, always written out as "Cohen's d". For a paired design it is $\bar{d}/s_d$.

Calculating *t* from the differences

To calculate t for a paired t-test, we use the one-sample t-test formula from the previous section, applied to the difference scores instead of raw scores.

To keep the arithmetic short, we first work through five of the 20 plots by hand. The next section runs the test on all 20. The table below shows the five plots and each step toward t.

Table 3: Soil moisture (%) before and after restoration at five plots, with the steps for the paired t.

Plot	Before	After	Difference	Deviation from mean difference score	Squared deviation
1	18	21	3	0.2	0.04
2	22	26	4	1.2	1.44
3	16	22	6	3.2	10.24
4	25	29	4	1.2	1.44
5	21	18	-3	-5.8	33.64
Sums	102	116	14	0	46.8
Means	20.4	23.2	2.8	0	9.36
				sd	3.421
				SE	1.53
				t	1.83

The five differences are 3, 4, 6, 4, and −3. They add up to 14, and their squared deviations from 2.8 add up to 46.8.

$$\bar{d} = 14/5 = 2.8 \text{ percentage points}$$

$$s_d = \sqrt{46.8/4} = 3.42$$

$$SE_{\bar{d}} = 3.42/\sqrt{5} = 1.53$$

$$t = 2.8/1.53 = 1.83$$

R gives almost the same numbers. The table rounds at each step, so its values differ slightly from R's.

Table 4: Paired t-test on 5 restoration plots.

Mean difference score	t	df	p
2.8	1.83	4	0.141

The mean difference score was +2.8 percentage points, so soil moisture increased on average. Getting a p-value from this t, and choosing one-tailed vs. two-tailed, works exactly like it did for the one-sample case (Section). Two-tailed, $t(4) = 1.83$, $p = 0.141$, which is not significant. With only five plots, a mean increase this large would not be unusual even if restoration had no effect on soil moisture.

Scaling to the full sample

The five plots above are a subset of the full restoration study, which measured all 20 paired plots. In R, give `t.test()` the two columns and set `paired = TRUE`. It subtracts the second column from the first, so this call tests After minus Before. Running `t.test(after_full - before_full, mu = 0)` on the differences gives the same result.

```
t.test(after_full, before_full, paired = TRUE)
```

Table 5: Paired t-test on all 20 restoration plots.

Mean difference score	t	df	p	95% CI lower	95% CI upper
1.9	2.5	19	0.022	0.31	3.49

At $n = 20$, $t(19) = 2.50$, $p = 0.022$, a significant result. The mean difference score (1.90 percentage points) is smaller than the 5-plot subset's 2.8. The larger t comes from the standard error, which fell from 1.53 to 0.76 with four times as many plots. Four times as many plots doubles $\sqrt{n}$, from $\sqrt{5} = 2.24$ to $\sqrt{20} = 4.47$, which halves the standard error. The SD of the differences barely changed, from 3.42 to 3.4. A mean increase of 1.9 points would be unlikely if restoration had no effect. With 20 plots, the test had more power to detect an effect of this size.

Analyzed wrongly as two independent groups, the same 20 plots give $t(38) = 1.34$, $p = 0.187$, and the test fails to reject H_0. The pooled SD still contains the plot-to-plot differences that pairing removes.

Confidence interval. The CI for a paired test follows the same recipe as the one-sample CI, the estimate plus or minus t_{crit} times the estimate's standard error. Here the estimate is $\bar{d}$, its standard error is $s_d/\sqrt{n}$, and t_{crit} comes from the t distribution with $df = 19$:

$$\bar{d} \pm t_{crit} \cdot \frac{s_d}{\sqrt{n}} = 1.90 \pm 2.09 \times 0.76 = [0.31, 3.49]$$

This is the interval R reports in the table above. It is the 95% CI for the effect, the mean difference score for soil moisture, in percentage points.

Table 6 summarizes the paired t-test.

Table 6: Test card: paired t-test.

Purpose	Did the units change, on average? Same units measured twice
Hypotheses	$H_0 : \mu_d = 0$, where μ_d is the population mean of the difference scores $H_a : \mu_d \neq 0$
Assumptions	Difference scores approximately normal
Test statistic	$d_i = \text{After}_i - \text{Before}_i$ $t = \bar{d}/(s_d/\sqrt{n})$ $df = n - 1$, where n is the number of pairs
Report	$\bar{d}$ and 95% CI $t(df)$ and p Optionally, Cohen's $d = \bar{d}/s_d$

The independent-samples *t*-test

The independent-samples t-test is the third and last design in this chapter.

A **between-subjects** design has different units in each condition, and each unit is measured once. The independent-samples t-test is the t-test for this design. In the example below, two groups of 5 sites make 10 units in total, none of them paired. Without repeated measures on the same unit, there's no meaningful way to subtract scores between conditions the way the paired t-test does.

The numerator is the mean difference between groups, as in the other two tests. The denominator differs from the paired case, because now we need a single estimate of variability built from two separate samples instead of one.

A natural first guess is to average the two groups' standard errors. Here $SE_A = s_A/\sqrt{n_A}$ is the standard error of Group A's mean, and SE_B is the same for Group B.

$$t = \frac{\bar{x}_A - \bar{x}_B}{\left(\frac{SE_A + SE_B}{2}\right)}$$

That average understates how much a difference between two means varies. The standard error of a difference between two independent means is $\sqrt{SE_A^2 + SE_B^2}$, because variances add when two independent quantities are subtracted. This is always larger than the average of the two SEs. The pooled independent-samples t-test also assumes the two populations share one variance. Both samples estimate that shared variance, so the test combines them, which gives a better estimate, with more degrees of freedom, than either group alone. The **pooled variance**, s_p^2, is that combined estimate, a weighted average of the two sample variances, s_A^2 and s_B^2, with each weighted by its degrees of freedom (the weighted mean, Section 1.9.5). The **pooled standard deviation**, s_p, is its square root. Putting s_p^2 in place of each group's variance in $\sqrt{SE_A^2 + SE_B^2}$ gives $\sqrt{s_p^2/n_A + s_p^2/n_B} = s_p\sqrt{1/n_A + 1/n_B}$, the denominator below:

$$t = \frac{\bar{x}_A - \bar{x}_B}{s_p\sqrt{\frac{1}{n_A} + \frac{1}{n_B}}}$$

where (note s_A^2 and s_B^2 here are variances, not standard deviations)

$$s_p = \sqrt{\frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}}$$

The pooled test has $df = n_A + n_B - 2$, one degree of freedom used for each group's mean. With two groups of 5, that is $df = 8$, the 8 in $t(8)$ below.

The pooled test adds two test-specific assumptions: the outcome is approximately normal within each group, and the two groups have similar variances. When the spreads clearly differ, use Welch's version instead, which does not pool the variances.

Table 7 compares the pooled test with Welch's version.

Table 7: Pooled and Welch's independent-samples t-tests.

	Pooled	Welch's
Assumes similar spread in both groups?	Yes	No
Standard error (denominator)	$s_p\sqrt{1/n_A + 1/n_B}$, from one combined s_p	$\sqrt{SE_A^2 + SE_B^2}$, from each group's own SD
df	$n_A + n_B - 2$	Estimated by software, usually not a whole number
Where it appears	The formula in this chapter, and `t.test(a, b, var.equal = TRUE)`	R's `t.test()` default

The worked example compares bird species at ten field-edge sites. Five sites had their hedgerow removed (Group A), and five kept an intact hedgerow (Group B). At each site, one 10-minute point count recorded the number of bird species seen or heard. The counts are invented.

- **Null (H_0):** hedgerows make no difference to the mean number of bird species, $\mu_A = \mu_B$.
- **Alternative (H_a):** the mean number of species differs, $\mu_A \neq \mu_B$.

The code below follows the formula step by step in R, then checks it against `t.test()`.

```
## By "hand" using R
# Bird species per point count
a <- c(1,2,3,4,5) # Group A, hedgerow removed
b <- c(3,5,4,7,9) # Group B, hedgerow intact
n_a <- length(a) # sample size for A
n_b <- length(b) # sample size for B

mean_difference <- mean(a)-mean(b) # compute mean difference

variance_a <- var(a) # compute variance for A
variance_b <- var(b) # compute variance for B

# Compute top part and bottom part of sp formula

sp_numerator <- (n_a - 1) * variance_a + (n_b - 1) * variance_b
sp_denominator <- n_a + n_b - 2
sp <- sqrt(sp_numerator/sp_denominator) # compute sp


# compute t following the formula

t <- mean_difference / (sp * sqrt(1/n_a + 1/n_b))
```

Group A's counts (1, 2, 3, 4, 5) have mean 3.0 and variance 2.5. Group B's counts (3, 5, 4, 7, 9) have mean 5.6 and variance 5.8. The pooled variance, the standard error of the difference, and t follow:

$$s_p^2 = (4 \times 2.5 + 4 \times 5.8)/8 = 4.15$$

$$s_p = \sqrt{4.15} = 2.04$$

$$SE = 2.04\sqrt{1/5 + 1/5} = 1.29$$

$$t = (3.0 - 5.6)/1.29 = -2.02$$

```
t.test(a, b, var.equal = TRUE)
```

Table 8: Independent-samples t-test (pooled) on bird species, Group A minus Group B.

Mean difference	t	df	p	95% CI lower	95% CI upper
-2.6	-2.02	8	0.078	-5.57	0.37

The t computed from the formula is -2.02, and the t from `t.test()` in the table matches it. R also reports $df = 8$, $p = 0.078$, and the 95% confidence interval for the difference. The p-value is above 0.05, so we fail to reject H_0.

Table 9 shows that reversing the order of subtraction flips the sign of the difference and of t and leaves p unchanged.

Table 9: The bird species data subtracted in each order.

Group 1 − Group 2	Difference in means	t	Reads as
A − B (removed minus intact)	$3.0 - 5.6 = -2.6$	-2.02	Sites without a hedgerow averaged fewer species
B − A (intact minus removed)	$5.6 - 3.0 = +2.6$	$+2.02$	Sites with a hedgerow averaged more species

Confidence interval. The recipe is the same as before, the estimate plus or minus t_{crit} times its standard error. The estimate is $\bar{x}_A - \bar{x}_B$, and its standard error, $SE_{\text{diff}} = s_p\sqrt{1/n_A + 1/n_B}$, is the denominator of the t just computed. With t_{crit} from the t distribution with $df = 8$:

$$(\bar{x}_A - \bar{x}_B) \pm t_{crit} \cdot SE_{\text{diff}} = -2.60 \pm 2.31 \times 1.29 = [-5.57, 0.37]$$

This is the interval R reports in the table above. It is the 95% CI for the effect, the mean difference in bird species between the two groups.

Cohen's d for two independent groups divides the mean difference by the pooled standard deviation, Cohen's $d = (\bar{x}_A - \bar{x}_B)/s_p$. Here $s_p = 2.04$, so Cohen's $d = -2.6/2.04 = -1.28$. Sites without a hedgerow averaged 1.28 pooled standard deviations fewer bird species than sites with one.

Table 10 summarizes the independent-samples t-test.

Table 10: Test card: independent-samples t-test.

Purpose	Do two separate groups differ on average? Different units in each group
Hypotheses	$H_0 : \mu_A = \mu_B$ $H_a : \mu_A \neq \mu_B$
Assumptions	Approximately normal within each group Similar spread in both groups (pooled version) When the spreads clearly differ, use Welch's version
Test statistic	$t = \dfrac{\bar{x}_A - \bar{x}_B}{s_p\sqrt{\dfrac{1}{n_A} + \dfrac{1}{n_B}}}$ $s_p = \sqrt{\dfrac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}}$ $df = n_A + n_B - 2$
Report	$\bar{x}_A - \bar{x}_B$ and 95% CI $t(df)$ and p Optionally, Cohen's $d = (\bar{x}_A - \bar{x}_B)/s_p$

Table 11 sets the three forms side by side. Each one divides an effect by its standard error. The design decides what counts as the effect and how its standard error is estimated.

Table 11: The three t-tests compared.

	One-sample	Paired	Independent-samples
Design	One sample against a reference value	The same units measured twice	Two separate groups
Effect (numerator, and center of the 95% CI)	$\bar{x} - \mu_0$	$\bar{d}$	$\bar{x}_A - \bar{x}_B$
Standard error (denominator)	$s/\sqrt{n}$	$s_d/\sqrt{n}$	$s_p\sqrt{1/n_A + 1/n_B}$
df	$n - 1$	$n - 1$ (n = number of pairs)	$n_A + n_B - 2$
Rejects H_0 (two-tailed, $\alpha = 0.05$) when	μ_0 is outside the 95% CI	0 is outside the 95% CI	0 is outside the 95% CI
R call	`t.test(x, mu = 50)`	`t.test(after, before, paired = TRUE)`	`t.test(a, b, var.equal = TRUE)`

! Results reporting: t-tests

All three forms report the same things in the same order. Each write-up gives the effect and its direction in plain language, the test statistic with its df, p, and the 95% CI for the effect. A write-up can add Cohen's d, as the independent-samples example does.
One-sample, where the effect is the sample mean against the reference value:

> Mean sound level near the highway was 61.0 dB, which did not differ significantly from the 50 dB guideline, $t(9) = 1.94$, $p = 0.084$, 95% CI [48.18, 73.82] dB.

Paired, where the effect is the mean *difference* score:

> Soil moisture increased by 1.90 percentage points after restoration, $t(19) = 2.50$, $p = 0.022$, 95% CI [0.31, 3.49] percentage points.

Independent-samples, where the effect is the difference between two group means:

> Bird species richness did not differ significantly between sites without a hedgerow and sites with one, mean difference −2.60 species, $t(8) = -2.02$, $p = 0.078$, 95% CI [-5.57, 0.37] species, Cohen's $d = -1.28$.

Note the paired write-up never mentions the before and after means separately. The difference score *is* the unit of analysis, so that single number plus its CI is the whole result.

Simulating data for *t*-tests

Earlier in this chapter we simulated t under H_0 and saw its null distribution in Figure 3. The same kind of simulation also gives a p-value for each repetition, and those p-values show what a test does when H_0 is true and when it is false.

Simulating under conditions you choose shows how sampling error affects results for a given sample size, expected difference, and amount of variation. Table 5.3 shows power estimated the same way, with each estimate checked against the exact power computed from the t distribution.

Simulating a one-sample *t*-test

A simulation has two steps.

1. Decide what the population looks like. Here each sample is 30 sites, with values drawn from a normal distribution with mean 50 and SD 25.
2. Draw one sample from that population, run the t-test, and save the statistics you care about (t and p). Repeat 1,000 times and look at how they are distributed.

Every test compares the sample mean to $\mu_0 = 50$, which is also the true population mean, so H_0 is true in every repetition. Figure 9 shows the 1,000 p-values it produces.

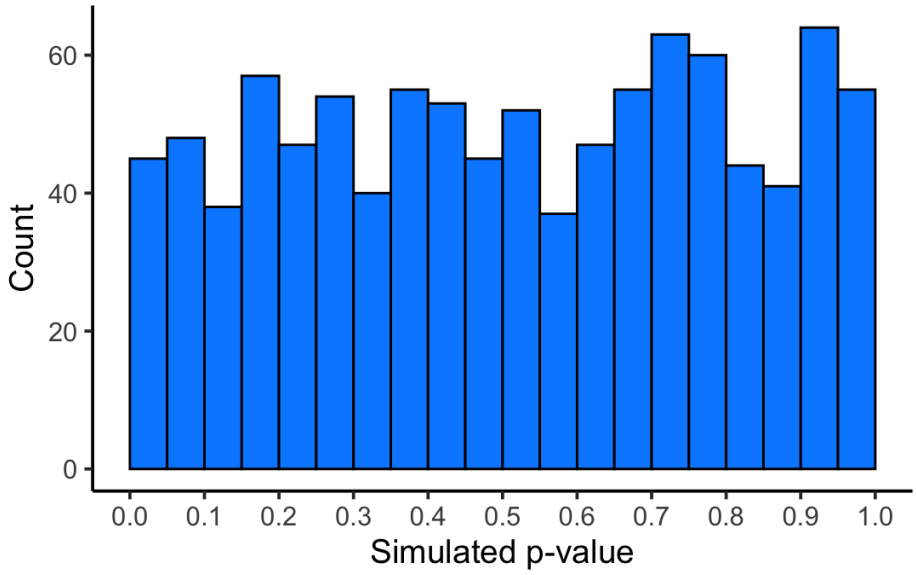


Figure 9: 1,000 simulated p-values from samples of 30 when H_0 is true. The distribution is flat.

The p-values in Figure 9 are flat, as they should be when H_0 is true. Under H_0, the critical values are set so that the test rejects at $\alpha = 0.05$ in 5% of samples, at $\alpha = 0.10$ in 10% of samples, and so on for any α. A p-value falls below 0.05 in 5% of tests and below 0.10 in 10% of tests, so every range of p-values 0.05 wide, such as 0 to 0.05 or 0.90 to 0.95, holds about 5% of the tests.

The first bar holds the 45 p-values below 0.05. Each one is a test that rejected a true H_0, a Type I error. Together they make up about 5% of the 1,000, which is the α we chose.

Simulating paired and independent-samples tests

The paired and independent-samples cases behave the same way, because they use the same t calculation on different inputs. A paired test runs a one-sample t on the difference scores. An independent-samples test uses the pooled standard error in the denominator. Simulate either one under H_0 and you get the same two pictures: a t distribution centered on zero, and a flat p distribution.

Section 5.4.1 simulated 1,000 experiments with a real difference of 2.5 points (two groups of 10, SD 5, so Cohen's $d = 0.5$) and counted how often $p < 0.05$ to estimate power. Figure 10 shows the 1,000 p-values from the same design, with the group means set at 10 and 12.5.

With a real difference, the p-values pile up near 0, while under H_0 in Figure 9 they spread evenly from 0 to 1. Here H_0 is false, so rejecting it is a correct decision, not a Type I error. The first bar holds the 187

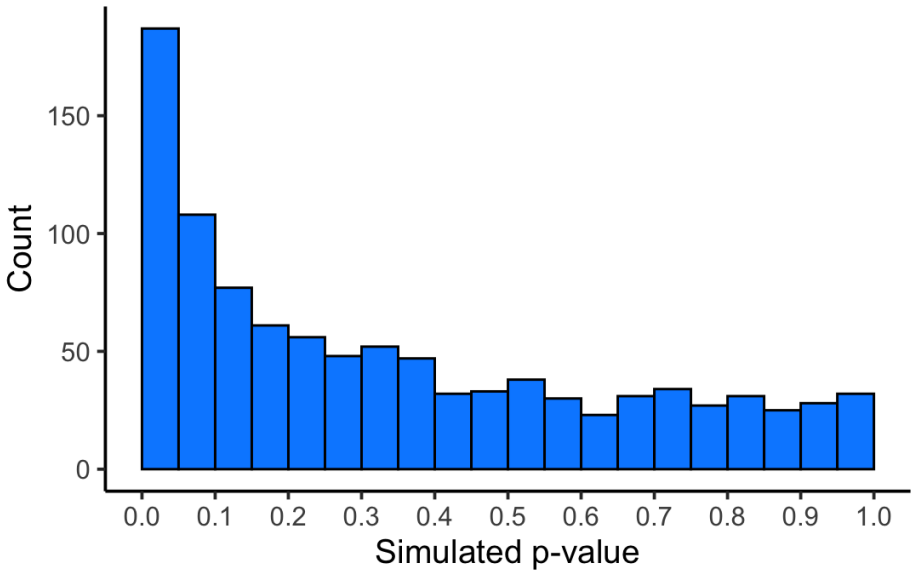


Figure 10: 1,000 simulated p-values ($n = 10$ per group) when there is a true difference.

experiments that rejected H_0, and their share of the 1,000, 18.7%, estimates the design's power. The exact power for this design, computed from the t distribution, is 19%. The other 813 missed a difference that was really there. Each of these is a Type II error (Chapter 5). Chapter 5 showed how a larger sample raises power.

Chapter Summary

Why it matters. The t-test measures a signal against the noise around it. ANOVA, regression, and the rest of the course use the same logic.

Core ideas

- **Signal over noise.** A difference is only impressive relative to the variability around it. Every t statistic is a ratio: the effect you observed on top, an estimate of how much that effect could vary by chance on the bottom.
- **Three forms, one idea.** The one-sample, paired, and independent-samples t-tests differ in how they compute the effect and its standard error, but all three ask whether an observed difference is large compared to its own standard error. The paired test is the one-sample test applied to difference scores.
- **Why t and not z.** When the population SD σ is unknown, we estimate it with s, which adds uncertainty. The t distribution accounts for that with heavier tails than the normal, and it approaches the normal as degrees of freedom grow.
- **Assumptions come in two kinds.** *Independence and a sound study design* apply to every test in this book and are assumed from here on. The *test-specific* assumption is approximate normality: of the outcome (one-sample), of the difference scores (paired), or of each group (independent samples). The pooled independent-samples test also assumes similar variances.
- **How worried to be depends on n.** Thanks to the central limit theorem, a t-test's p-values stay close to correct with non-normal data once samples are reasonably large. t-tests are least trustworthy when samples are small *and* skewed, which describes a lot of environmental field data. Plot your data first, and treat a Shapiro-Wilk result as evidence rather than a gate, since it has the least power with small samples, when a violation matters most.

- **The tail choice changes p for the same data.** For the highway noise example, $t(9) = 1.94$ gives a one-tailed p of 0.042 and a two-tailed p of 0.084. Decide which you need before seeing the result, using the rule from Sections [-4.5.1] and [-5.2.1].

Student, A. 1908. "The Probable Error of a Mean." *Biometrika* 6: 1–2.

Welch, B. L. 1947. "The Generalization of 'Student's' Problem When Several Different Population Variances Are Involved." *Biometrika* 34: 28–35.